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# 1 Some Data-Cleaning Tools

- reading: @WickhamTidyData2014

## 1.1 Rectangular and non-rectangular data

Figure 1: Tweet by Nathan Oseroff-Spicer (nathanoseroff): 'It makes sense, though, since there are rows and columns. It's scientific'. Source: <https://twitter.com/nathanoseroff/status/1321565407568867332>

- Data represented in rows and columns is called *rectangular data*.
- Common non-rectangular forms include XML and JSON
  - Sometimes you might get data in the form of a PDF or image. This is called *cursèd data*.

## 1.2 Tidy Data

- conventions for organizing data for analysis
  1. Each variable forms a column.
  2. Each observation forms a row.
  3. Each type of observational unit forms a table.
- Example: student demographics and major, for students who took a philosophy course at UC Davis, 2005-2015

- slightly rephrased
  - 1 table = 1 unit of observation
    - student-term (major)
    - student (demographics)
  - 1 row = 1 unique observed unit
    - 1 row = 1 student-term combination
    - 1 row = 1 student
  - Every column is either an *identifier* or a *measured variable*
    - rule of thumb: 1 row = 1 unique combination of identifiers
    - major table
      - identifiers: student ID, `term`
      - measured variables: admit term, relative term, major
      - NB the `term` identifier is constructed from two measured variables
    - demographics table

| id | admit_term | rel_term | term | major |
|---|---|---|---|---|
| 000[redacted]B68 | 201010 | YR1FALL | 201010 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR1WINTER | 201101 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR1SPRING | 201103 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR1SUMMER_I | 201105 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR1SUMMER_II | 201107 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR2FALL | 201110 | LPOL-Political Science |
| 000[redacted]B68 | 201010 | YR2WINTER | 201201 | LPOL-Political Science |

| id | gender | ethnicity_code | ethnicity | race | first_gen | low_income | admission_type | ever_phil | poc | race4 | demographic | major_at_first_phil |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8F[redacted]73 | F | OT | Other | Other | FALSE | FALSE | Transfer | FALSE | FALSE | Other | F.Other | LENL-English |
| 0F[redacted]99C | M | (Missing) | Other | Other | TRUE | FALSE | Transfer | FALSE | FALSE | Other | M.Other | APME-Pre-Managerial Economics |
| 1C[redacted]9C2 | F | WH | White/Caucasian | White | TRUE | FALSE | Transfer | FALSE | FALSE | White | F.White | LPSC-Psychology |
| 3E[redacted]BEA | F | MX | Mexican-Am/Mexican/Chicano | Hispanic | TRUE | FALSE | Transfer | TRUE | TRUE | BHIP | F.BHIP | LPHI-Philosophy |
| 95[redacted]693 | F | KO | Korean-American/Korean | Asian | TRUE | TRUE | Transfer | FALSE | TRUE | Asian | F.Asian | AHDE-Human Development |
| A1[redacted]6B0 | M | CH | Chinese-American/Chinese | Asian | FALSE | TRUE | Freshman | FALSE | TRUE | Asian | M.Asian | EMEC-Mechanical Engineering |
| 6C[redacted]127 | M | FP | Filipino/Filipino-American | Asian | FALSE | FALSE | Transfer | FALSE | TRUE | Asian | M.Asian | ABIS-Biological Sciences |

Figure 2: Two tables in a tidy dataset. The first table records students' majors during each term they were enrolled; the second (next figure) records demographics. The tables have different units of observation, but can be linked using the ID (internal to this dataset) for an analysis at the level of individual students. Other tables in this dataset include final grades for philosophy courses taken by the students, and demographics of the instructors of those courses.

- identifier: student ID
- measured variables: gender, race and ethnicity variables, first gen status, low income status, admission type, whether they ever majored in philosophy, major the semester they took their first philosophy course
- NB here "measured variable" includes both variables in the raw data as well as variables I constructed for analysis purposes (`poc`, `demographic`)

## 1.3 Tools for tidying data

```r
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.0 --

## v ggplot2 3.3.2     v purrr   0.3.4
## v tibble  3.0.4     v dplyr   1.0.2
## v tidyr   1.1.2     v stringr 1.4.0
## v readr   1.4.0     v forcats 0.5.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```r
theme_set(theme_minimal())

data_folder = file.path('data')
data_zip = file.path(data_folder, 'tidy.zip')
## Get the data
if (!file.exists(data_zip)) {
    download.file('https://github.com/data-science-methods/book/raw/master/data/tidy.zip',
                  data_zip)
}
unzip(data_zip, overwrite = FALSE, exdir = data_folder)
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
```

```
## overwriting file 'data/billboard.csv
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/weather.csv
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/tb.sav
```

```
## Warning in unzip(data_zip, overwrite = FALSE, exdir = data_folder): not
## overwriting file 'data/pew.csv
```

### 1.3.1 3.1: The Pew Dataset

- pivot_longer()
- if_else()
- fct_inorder()

```
pew_df = read_csv(file.path(data_folder, 'pew.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
## -- Column specification --------------------------------------------------------
## cols(
##   X1 = col_double(),
##   religion = col_character(),
##   `<$10k` = col_double(),
##   `$10-20k` = col_double(),
##   `$20-30k` = col_double(),
##   `$30-40k` = col_double(),
##   `$40-50k` = col_double(),
##   `$50-75k` = col_double(),
##   `$75-100k` = col_double(),
##   `$100-150k` = col_double(),
##   `>150k` = col_double(),
##   `Don't know/refused` = col_double()
## )
```

```
head(pew_df)
```

| X1 | religion | <$10k | $10-20k | $20-30k | $30-40k | $40-50k | $50-75k | $75-100k | $100-150k | >150k |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Agnostic | 27 | 34 | 60 | 81 | 76 | 137 | 122 | 109 | 84 |
| 2 | Atheist | 12 | 27 | 37 | 52 | 35 | 70 | 73 | 59 | 74 |
| 3 | Buddhist | 27 | 21 | 30 | 34 | 33 | 58 | 62 | 39 | 53 |
| 4 | Catholic | 418 | 617 | 732 | 670 | 638 | 1116 | 949 | 792 | 633 |
| 5 | Don't know/refused | 15 | 14 | 15 | 11 | 10 | 35 | 21 | 17 | 18 |
| 6 | Evangelical Prot | 575 | 869 | 1064 | 982 | 881 | 1486 | 949 | 723 | 414 |

```
pew_clean = pew_df %>%
    ## Drop the row number column
    select(-X1) %>%
    ## Lengthen: 1 row per religion x income level
    pivot_longer(cols = `<$10k`:`Don't know/refused`,
                 names_to = 'income',
                 values_to = 'freq') %>%
    ## Cleaning `income`:
    ## - Make "Don't know/refused" into NA
    ## - Coerce to ordered factor
```

```r
    mutate(income = if_else(str_detect(income, "Don't know"),
                            NA_character_,
                            income),
           income = fct_inorder(income))
head(pew_clean)
```

| religion | income | freq |
|---|---|---|
| Agnostic | <$10k | 27 |
| Agnostic | $10-20k | 34 |
| Agnostic | $20-30k | 60 |
| Agnostic | $30-40k | 81 |
| Agnostic | $40-50k | 76 |
| Agnostic | $50-75k | 137 |

```r
ggplot(pew_clean, aes(religion, freq, fill = fct_rev(income))) +
    geom_col() +
    scale_fill_viridis_d(na.value = 'grey50') +
    coord_flip()
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
```

```
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

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## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
```

```
## <80>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

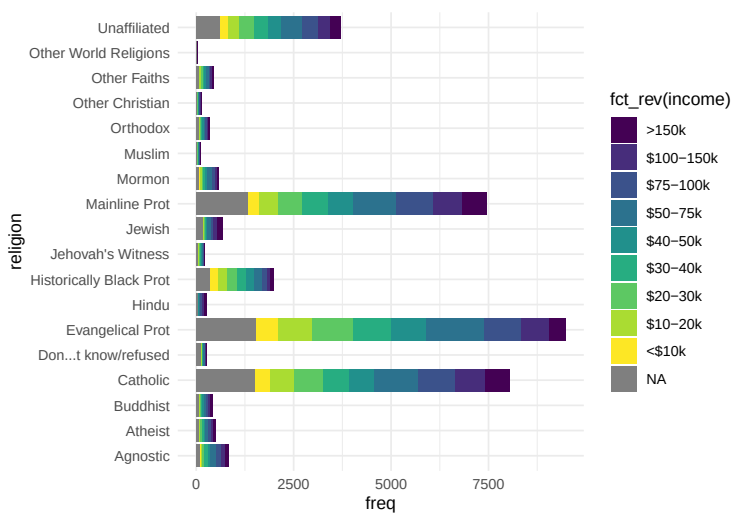
```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <e2>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <80>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Don’t know/refused' in 'mbcsToSbcs': dot substituted for
## <99>
```




### 1.3.2 3.1: The Billboard Dataset

- A little helper function for parsing
- `str_remove()`
- Coercion
- Date arithmetic

```
billboard_df = read_csv(file.path(data_folder, 'billboard.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
```

```
## -- Column specification --------------------------------------------------------
## cols(
##   .default = col_double(),
##   artist = col_character(),
##   track = col_character(),
##   time = col_time(format = ""),
##   date.entered = col_date(format = ""),
##   wk66 = col_logical(),
##   wk67 = col_logical(),
##   wk68 = col_logical(),
##   wk69 = col_logical(),
##   wk70 = col_logical(),
##   wk71 = col_logical(),
##   wk72 = col_logical(),
##   wk73 = col_logical(),
##   wk74 = col_logical(),
##   wk75 = col_logical(),
##   wk76 = col_logical()
## )
## i Use `spec()` for the full column specifications.
```

```r
head(billboard_df)
```

| X1 | year | artist | track | time | date.entered | wk1 | wk2 | wk3 | wk4 | wk5 | wk6 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2000 | 2 Pac | Baby Don't Cry | 04:22:00 | 2000-02-26 | 87 | 82 | 72 | 77 | 87 | 94 |
| 2 | 2000 | 2Ge+her | The Hardest Part Of ... | 03:15:00 | 2000-09-02 | 91 | 87 | 92 | NA | NA | NA |
| 3 | 2000 | 3 Doors Down | Kryptonite | 03:53:00 | 2000-04-08 | 81 | 70 | 68 | 67 | 66 | 57 |
| 4 | 2000 | 3 Doors Down | Loser | 04:24:00 | 2000-10-21 | 76 | 76 | 72 | 69 | 67 | 65 |
| 5 | 2000 | 504 Boyz | Wobble Wobble | 03:35:00 | 2000-04-15 | 57 | 34 | 25 | 17 | 17 | 31 |
| 6 | 2000 | 98^0 | Give Me Just One Nig... | 03:24:00 | 2000-08-19 | 51 | 39 | 34 | 26 | 26 | 19 |

```r
#' Helper function to parse weeks to integers in the Billboard dataset
#' @param wk_chr Vector of week characters, with the pattern `c('wk1', 'wk2')`
#' @return Vector of integers, eg, `c(1L, 2L)`
parse_week = function(wk_chr) {
    wk_int = wk_chr %>%
        str_remove('wk') %>%
        as.integer()
    return(wk_int)
}

billboard_clean = billboard_df %>%
    ## Drop the row number column
    select(-X1) %>%
    ## Pivot week down and remove NA rows
    pivot_longer(cols = starts_with('wk'),
                 names_to = 'week',
                 values_to = 'rank') %>%
    filter(!is.na(rank)) %>%
    ## Parse week and calculate date
    ## NB Base R can handle date arithmetic
    ## For month arithmetic see ?lubridate::`%m+%`
    mutate(week = parse_week(week),
           date = date.entered + (week-1)*7) %>%
    ## Clean columns
```

```
    select(year, artist, time, track, date, week, rank)
head(billboard_clean)
```

| year | artist | time | track | date | week | rank |
|---|---|---|---|---|---|---|
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-02-26 | 1 | 87 |
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-03-04 | 2 | 82 |
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-03-11 | 3 | 72 |
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-03-18 | 4 | 77 |
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-03-25 | 5 | 87 |
| 2000 | 2 Pac | 04:22:00 | Baby Don't Cry | 2000-04-01 | 6 | 94 |

NB. Base R includes methods for doing date arithmetic, eg,

```
as.Date('2020-04-30') + 1
```

```
## [1] "2020-05-01"
```

If you need month arithmetic, check out the `lubridate` package, and specifically `%m+%` and related operators.

```
count(billboard_clean, artist, track) %>%
    arrange(desc(n)) %>%
    head()
```

| artist | track | n |
|---|---|---|
| Creed | Higher | 57 |
| Lonestar | Amazed | 55 |
| 3 Doors Down | Kryptonite | 53 |
| Hill, Faith | Breathe | 53 |
| Creed | With Arms Wide Open | 47 |
| Joe | I Wanna Know | 44 |

### 1.3.3 3.2: The TB Dataset

- `haven` package
- `separate()`
- `case_when()`

```
tb_df = haven::read_sav(file.path(data_folder, 'tb.sav'))
head(tb_df)
```

| country | year | m014 | m1524 | m2534 | m3544 | m4554 | m5564 | m65 | mu | f014 | f1524 | f2534 | f3544 | f4554 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AD | 2000 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA |
| AE | 2000 | 2 | 4 | 4 | 6 | 5 | 12 | 10 | NA | 3 | 16 | 1 | 3 | 0 |
| AF | 2000 | 52 | 228 | 183 | 149 | 129 | 94 | 80 | NA | 93 | 414 | 565 | 339 | 205 |
| AG | 2000 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | NA | 1 | 1 | 1 | 0 | 0 |
| AL | 2000 | 2 | 19 | 21 | 14 | 24 | 19 | 16 | NA | 3 | 11 | 10 | 8 | 8 |
| AM | 2000 | 2 | 152 | 130 | 131 | 63 | 26 | 21 | NA | 1 | 24 | 27 | 24 | 8 |

```
tb_clean = tb_df %>%
    ## One row per observation = country x gender x age group
    pivot_longer(m014:fu,
                 names_to = 'column',
                 values_to = 'cases') %>%
    ## Separate gender and age
    separate(column, into = c('gender', 'age'),
             sep = 1) %>%
    ## Wrangle age:
    ## - Split into upper and lower bounds
```

```r
    ## - Recombine with -
    ## - Fix missing age and 65+
    ## - Set order
    ## - Clean up the intermediate columns for upper and lower bounds
    separate(age, into = c('age_lower', 'age_upper'),
             sep = -2) %>%
    mutate(age = str_c(age_lower, '-', age_upper),
           age = case_when(age == '-u'  ~ NA_character_,
                           age == '-65' ~ '65+',
                           TRUE         ~ age),
           age = fct_inorder(age)) %>%
    select(-age_lower, -age_upper) %>%
    ## Put in order, with ID columns first and then the variable
    select(country, year, gender, age, cases)
head(tb_clean)
```

| country | year | gender | age | cases |
|---|---|---|---|---|
| AD | 2000 | m | 0-14 | 0 |
| AD | 2000 | m | 15-24 | 0 |
| AD | 2000 | m | 25-34 | 1 |
| AD | 2000 | m | 35-44 | 0 |
| AD | 2000 | m | 45-54 | 0 |
| AD | 2000 | m | 55-64 | 0 |

### 1.3.4 3.3: The Weather Dataset

- `pivot_wider()`
- `str_remove()`

```r
weather_df = read_csv(file.path(data_folder, 'weather.csv'))
```

```
## Warning: Missing column names filled in: 'X1' [1]
```

```
##
## -- Column specification --------------------------------------------------------
## cols(
##   .default = col_double(),
##   id = col_character(),
##   element = col_character(),
##   d9 = col_logical(),
##   d12 = col_logical(),
##   d18 = col_logical(),
##   d19 = col_logical(),
##   d20 = col_logical(),
##   d21 = col_logical(),
##   d22 = col_logical(),
##   d24 = col_logical()
## )
## i Use `spec()` for the full column specifications.
```

```r
head(weather_df)
```

| X1 | id | year | month | element | d1 | d2 | d3 | d4 | d5 | d6 | d7 | d8 | d9 | d10 | d11 | d12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | MX17004 | 2010 | 1 | tmax | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 2 | MX17004 | 2010 | 1 | tmin | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 3 | MX17004 | 2010 | 2 | tmax | NA | 27.3 | 24.1 | NA | NA | NA | NA | NA | NA | NA | 29.7 | NA |
| 4 | MX17004 | 2010 | 2 | tmin | NA | 14.4 | 14.4 | NA | NA | NA | NA | NA | NA | NA | 13.4 | NA |
| 5 | MX17004 | 2010 | 3 | tmax | NA | NA | NA | NA | 32.1 | NA | NA | NA | NA | 34.5 | NA | NA |
| 6 | MX17004 | 2010 | 3 | tmin | NA | NA | NA | NA | 14.2 | NA | NA | NA | NA | 16.8 | NA | NA |

```r
weather_clean = weather_df %>%
    select(-X1) %>%
    ## Lengthen days
    pivot_longer(d1:d31,
                 names_to = 'day',
                 values_to = 'obs') %>%
    ## Widen `element`
    pivot_wider(names_from = element,
                values_from = obs) %>%
    ## Fix the class of `day`
    mutate(day = str_remove(day, 'd'),
           day = as.integer(day)) %>%
    ## Check against the paper by filtering to non-missing rows
    filter(!is.na(tmax) | !is.na(tmin))
head(weather_clean)
```

| id | year | month | day | tmax | tmin |
|---|---|---|---|---|---|
| MX17004 | 2010 | 1 | 30 | 27.8 | 14.5 |
| MX17004 | 2010 | 2 | 2 | 27.3 | 14.4 |
| MX17004 | 2010 | 2 | 3 | 24.1 | 14.4 |
| MX17004 | 2010 | 2 | 11 | 29.7 | 13.4 |
| MX17004 | 2010 | 2 | 23 | 29.9 | 10.7 |
| MX17004 | 2010 | 3 | 5 | 32.1 | 14.2 |